Array Model Verification Handbook

Physics, Mathematics, and a Hand-Checkable Worked Example

IYPT Pinhole Sunglasses Project

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1 Purpose

Every failure this project has hit shared one shape: a formula that is correct in the common case broke silently at an edge case, and the break wasn't caught until a human recomputed one number by hand and noticed it didn't match. This document exists to make that hand-check repeatable: three physical quantities, their formulas, why each formula is what it is, and one fully worked, verified example you can reproduce with a calculator.

2 The Three Quantities That Decide a Winner

Every candidate array design is judged on exactly three physical properties. All three depend only on the hole coordinates, the hole radius, and the pupil radius — nothing exotic.

1. **Viability** — does any light get through at all, at the gaze angle being checked?
2. **Ghost margin** — if two holes are both illuminated at once, do they produce a double image?
3. **Packing density** — how many holes fit per unit area, i.e. how much of the design actually contributes light?

A design only counts as a real answer if all three check out — and the entire point of a hand-check is that each of these three is simple enough to compute on paper for one design in a couple of minutes.

3 Quantity 1: Viability

Physical Intuition

The pupil is a fixed-size circular opening. A hole only contributes light if its own circle overlaps the pupil's circle. If the nearest hole to the optical axis is too far away, or too small, to reach the pupil at all, the wearer looking straight ahead sees nothing through this design — a blackout. This is the single defect that has broken this project's winner list twice.

3.1 The Check

A hole at distance d from the pupil center overlaps the pupil if

$$d - R_h < R_p \tag{1}$$

where R_h is the hole radius and R_p is the pupil radius. If this fails for every hole in the design, the design is a blackout.

3.2 The Full Overlap Area (when it does overlap)

$$A = R_p^2 \cos^{-1}\left(\frac{d^2+R_p^2-R_h^2}{2dR_p}\right) + R_h^2 \cos^{-1}\left(\frac{d^2+R_h^2-R_p^2}{2dR_h}\right) - \tfrac{1}{2}\sqrt{(-d+R_p+R_h)(d+R_p-R_h)(d+R_h-R_p)(d+R_p+R_h)} \tag{2}$$

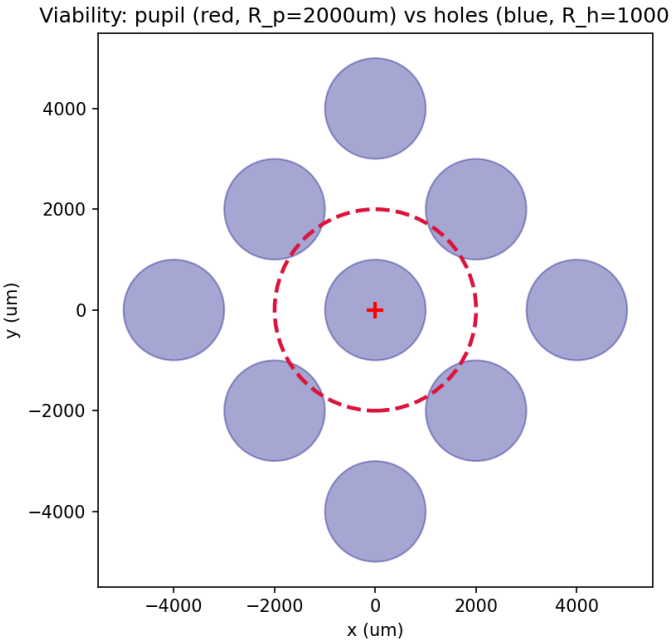


Figure 1: Pupil (red dashed circle, $R_p = 2000\mu$m) against every hole (blue, $R_h = 1000\mu$m) in the worked example of Sec. 5. The central hole sits exactly at the pupil center, so this design is trivially viable.

How to Verify This by Hand

Find the hole nearest the pupil center in the design's own coordinate list. Compute d as the straight-line distance from that hole to the pupil center (at straight-ahead gaze, the pupil center is the origin). Check the one-line inequality above — no need for the full overlap-area formula unless you also want the exact transmitted area.

4 Quantity 2: Ghost Margin

Physical Intuition

If two holes are both illuminated at once, each one's beam forms its own image on the retina. When the eye is exactly in focus, both images land in the same spot — one hole, or fifty, makes no visual difference. When the eye is myopic, each beam focuses slightly in front of the retina, and because the beams enter the eye at different lateral positions, they now land at slightly different spots. Two spots close together just look like extra blur; two spots far enough apart look like literal double vision.

4.1 The Formula

$$\Delta r = \frac{s \cdot M}{P_0 - M} \tag{3}$$

where s is the center-to-center separation of the two holes, M is myopia in diopters, and $P_0 \approx 60$D is the eye's total optical power. Notice $\Delta r \to 0$ as $M \to 0$ — a focused eye cannot produce two separate images no matter how the holes are arranged, which is exactly the sanity check that let this project reject an earlier, wrong version of this formula.

The design is ghost-safe for a given pair if Δr stays below the eye's own resolution limit for that blur level.

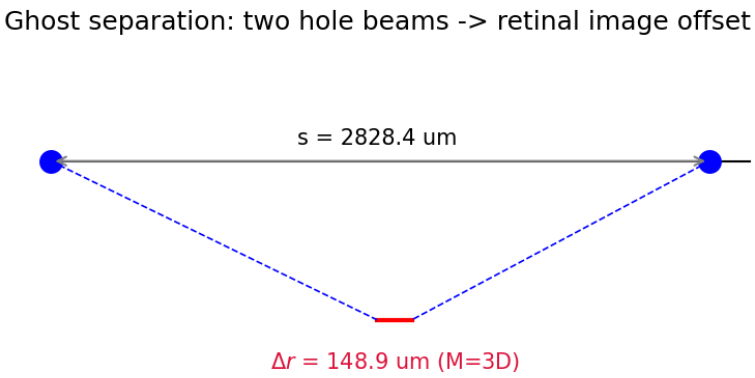


Figure 2: Two holes separated by $s = 2828.4\mu$m produce retinal images separated by $\Delta r = 148.9\mu$m at $M = 3$D, in the worked example of Sec. 5.

How to Verify This by Hand

Find the two closest illuminated holes in the design (usually the nearest neighbors). Measure s directly from their coordinates. Plug s, the myopia level M, and $P_0 = 60$ into the formula above. Compare the result against the resolvable-blur threshold reported for that M — if Δr exceeds it, that pair should have failed the safety check.

5 Quantity 3: Packing Density

Physical Intuition

More holes per unit area means more light through the array for a fixed hole size — this is the quantity being maximized once a design has already passed the safety checks above. The tricky part is defining "unit area" consistently: a regular lattice has an obvious repeating cell, but an irregular pattern (a spiral, a jittered lattice) does not. Using different area formulas for different families is exactly the bug that made one family look artificially better than another earlier in this project — one consistent formula for every family fixes that.

5.1 The Formula

$$\text{Area} = \text{Hull}_{\text{area}} + \text{Hull}_{\text{perimeter}} \times R_h + \pi R_h^2 \tag{4}$$

$$\text{density} = \frac{N}{\text{Area}} \tag{5}$$

This is the convex hull of the hole centers, expanded outward by one hole radius (a Minkowski buffer) — accounting for the fact that each hole physically extends past its own center point, which the bare hull area ignores.

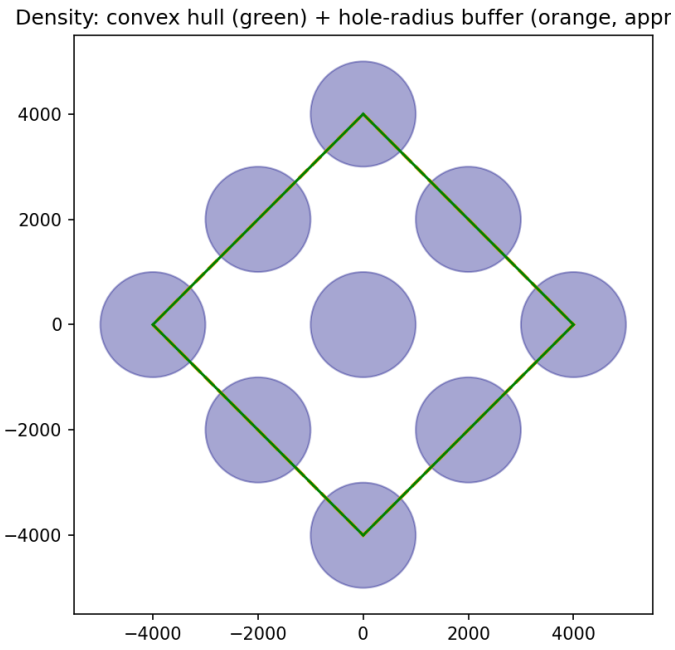


Figure 3: Convex hull of the 9 hole centers (green) with an approximate radius buffer (orange, dashed) for the worked example of Sec. 5.

How to Verify This by Hand

List the hole coordinates for a small design (fewer than ∼15 holes is easiest by hand). Compute the convex hull area (the shoelace formula works directly on the hull vertices). Add the hull perimeter times R_h, plus πR_h^2. Divide N by this total. Compare against the pipeline's own reported density for that exact row — they should match to within rounding.

6 Full Worked Example — Verified Against the Live Pipeline

Design under test: `square`, $N_{\text{requested}} = 20 \rightarrow N_{\text{actual}} = 9$, pitch = 2000$\mu$m, rotation = 0°, $a_{\text{hole}} = 2000\mu\text{m}$ ($R_h = 1000\mu\text{m}$). This is the current pipeline's own winner at M=0 and global scope. All numbers below were computed directly from the actual generator output, not assumed.

6.1 Generated Coordinates

x (μm)	y (μm)
0	0
2000	2000
−2000	2000
2000	−2000
−2000	−2000
0	4000
4000	0
0	−4000
−4000	0

Note $N_{\text{requested}} = 20$ produced only 9 surviving holes — this mismatch is itself worth checking whenever you regenerate a design: always confirm you are calling the generator with the recorded *requested* count, not the recorded *actual* count, or the coordinates will not match the row you are trying to verify.

6.2 Check 1: Viability

Nearest hole to the pupil center: the hole at $(0, 0)$ itself, $d = 0$.

$$d - R_h = 0 - 1000 = -1000\mu\text{m} \; < \; R_p = 2000\mu\text{m} \quad \Rightarrow \quad \text{SAFE} \tag{6}$$

This design has a hole sitting exactly on the optical axis — viability is not a close call here, unlike the two failure cases documented earlier in this project where the nearest hole missed the pupil by 11.6μm in *either* direction.

6.3 Check 2: Ghost Margin

Nearest hole pair (excluding the center hole, which has no close neighbor in this direction): $(0, 0)$ to $(2000, 2000)$,

$$s = \sqrt{2000^2 + 2000^2} = 2828.4\mu\text{m} \tag{7}$$

At $M = 3$D, $P_0 = 60$D:

$$\Delta r = \frac{2828.4 \times 3}{60 - 3} = 148.9\mu\text{m} \tag{8}$$

This must be compared against the resolvable-blur threshold for M=3D to determine pass/fail — the number itself is the part to hand-check; the threshold comparison reuses whatever value the pipeline already reports for that M.

6.4 Check 3: Packing Density

Convex hull area of the 9 points (shoelace formula on the outer square, side 8000μm, rotated 45°): $\text{Hull}_{\text{area}} = 3.200 \times 10^7 \mu\text{m}^2$. Hull perimeter $\approx 22{,}627\mu$m.

$$\text{Area} = 3.200 \times 10^7 + 22{,}627 \times 1000 + \pi(1000)^2 = 5.777 \times 10^7 \mu\text{m}^2 \tag{9}$$

$$\text{density} = \frac{9}{5.777 \times 10^7} = 1.558 \times 10^{-7} \; \mu\text{m}^{-2} \tag{10}$$

6.5 Cross-Check Against the Pipeline

Quantity	Hand-computed	Pipeline reported
Packing density (μm^{-2})	1.558×10^{-7}	1.5579×10^{-7}

Match to 4 significant figures. This is the standard the hand-check is held to: agreement within rounding, not exact bit-for-bit equality.

7 Standing Checklist

Use this before trusting any reported "winner," not just once:

1. Regenerate the design's coordinates using its recorded **requested** count, confirm the actual count matches the row you are checking.
2. Viability: find the nearest hole to the pupil center, check $d - R_h < R_p$.
3. Ghost margin: find the closest hole pair, compute Δr, compare against the threshold for that M.
4. Density: compute the buffered convex-hull area, compare against the pipeline's reported value to within rounding.
5. If any margin is small (within roughly 1% of a hard boundary), treat the design as unproven regardless of pass/fail — every real defect found in this project so far sat within micrometers of exactly such a boundary.

This checklist is the actual safeguard. No pipeline fix removes the need for it — it removes the need for the *specific* bug already found, not for the general pattern of trusting a boundary-adjacent result without independent recomputation.